(i) Total sum = 36, total sum of squares = 204. $\Longrightarrow$ Each subset's sum = 18, each subset's sum of squares = 102.

$$\boxed{A = \{1, 4, 6, 7\}, \qquad B = \{2, 3, 5, 8\}.}$$

(ii)

$$\text{LHS} = \sum a_k^3 + \sum \left(c^3 + 3c^2 b_k + 3c b_k^2 + b_k^3\right) = \sum a_k^3 + \sum b_k^3 + mc^3 + 3c^2 \sum b_k + 3c \sum b_k^2$$

$$\because \sum a_k = \sum b_k \text{ and } \sum a_k^2 = \sum b_k^2$$

$$\begin{aligned} \therefore \text{LHS} &= \sum a_k^3 + \sum b_k^3 + mc^3 + 3c^2 \sum a_k + 3c \sum a_k^2 \\ &= \sum_{k=1}^{m} b_k^3 + \sum_{k=1}^{m} (c + a_k)^3 \end{aligned}$$

(iii) From part (i), let $A' = \{1, 4, 6, 7\}, \quad B' = \{2, 3, 5, 8\}$
Then $\sum a'_k = \sum b'_k$ and $\sum {a'_k}^2 = \sum {b'_k}^2$.
When $c = 8$

$$\boxed{A = \{1, 4, 6, 7, 10, 11, 13, 16\}, \qquad B = \{2, 3, 5, 8, 9, 12, 14, 15\}.}$$

$$\sum_{x \in A} x = \sum a_k + \sum (8 + b_k), \qquad \sum_{x \in B} x = \sum b_k + \sum (8 + a_k),$$

$$\because \sum a'_k = \sum b'_k, \ \sum a_k = \sum b_k$$

$$\begin{aligned} \therefore \sum a_k^2 + \sum (c + b_k)^2 &= \sum a_k^2 + \sum (c^2 + 2c b_k + b_k^2) \\ &= \sum a_k^2 + mc^2 + 2c \sum b_k + \sum b_k^2 \\ &= \sum b_k^2 + mc^2 + 2c \sum a_k + \sum a_k^2 \\ &= \sum b_k^2 + \sum (c + a_k)^2 \end{aligned}$$

So, the sum of squares of A and B are equal.
By part (ii), $\sum a_k^3 + \sum (8 + b_k)^3 = \sum b_k^3 + \sum (8 + a_k)^3$
Thus, the sum of cubes of A and B are equal.

(iv) $A = \{1, 3, 4, 5, 9, 11\}$ and $B = \{0, 2, 6, 7, 8, 10\}$ form a balanced partition of $\{0, 1, 2, \ldots, 11\}$, and each subset contains 6 elements.
So $\{n + 1, n + 3, n + 4, n + 5, n + 9, n + 11\}$ and $\{n, n + 2, n + 6, n + 7, n + 8, n + 10\}$ form a balanced partition of $\{n, n + 1, n + 2, \ldots, n + 11\}$
Thus

$$\boxed{C = \{(n+1)^2, (n+3)^2, (n+4)^2, (n+5)^2, (n+9)^2, (n+11)^2\}} \text{ and}$$

$$\boxed{D = \{n^2, (n+2)^2, (n+6)^2, (n+7)^2, (n+8)^2, (n+10)^2\}}$$

is a pair of sets with the required property.

This solution was submitted by **Jon**

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Last updated: June 13, 2026